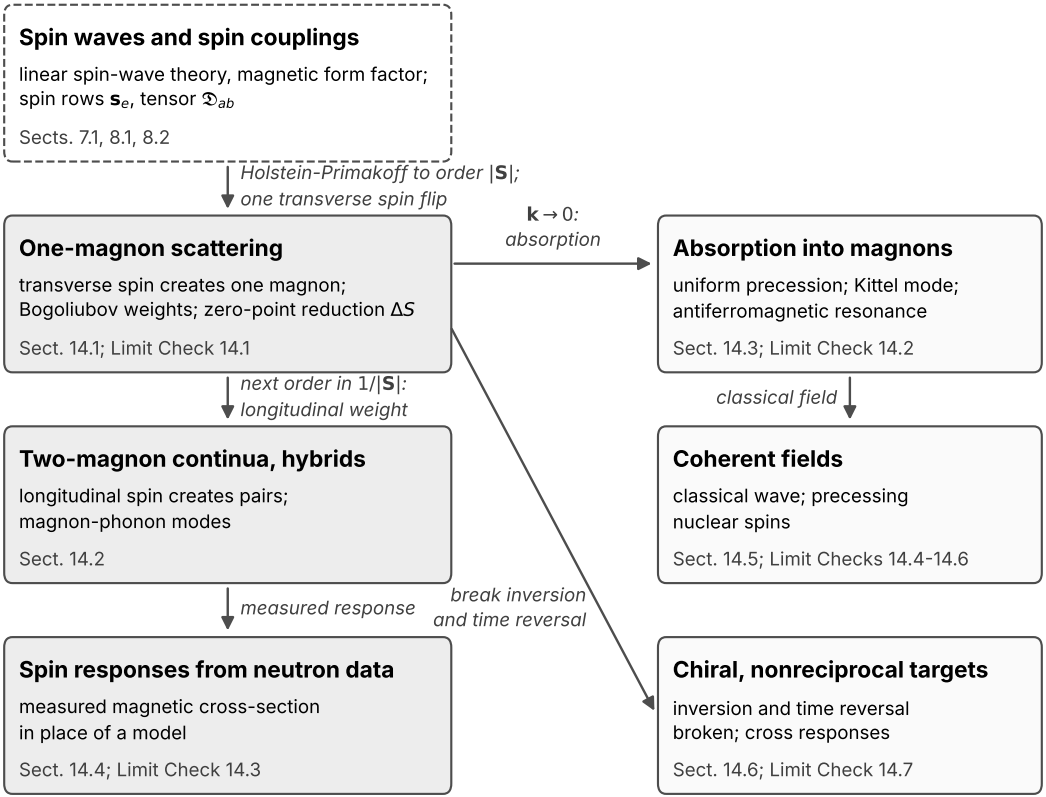
Spin waves and spin couplings
linear spin-wave theory, magnetic form factor; spin rows $\mathbf{s}_e$, tensor $\mathfrak{D}_{ab}$
Sects. 7.1, 8.1, 8.2
Holstein-Primakoff to order $|\mathbf{S}|$; one transverse spin flip
One-magnon scattering
transverse spin creates one magnon; Bogoliubov weights; zero-point reduction ΔS
Sect. 14.1; Limit Check 14.1
$\mathbf{k} \to 0$: absorption
Absorption into magnons
uniform precession; Kittel mode; antiferromagnetic resonance
Sect. 14.3; Limit Check 14.2
classical field
Coherent fields
classical wave; precessing nuclear spins
Sect. 14.5; Limit Checks 14.4-14.6
next order in $1/|\mathbf{S}|$: longitudinal weight
Two-magnon continua, hybrids
longitudinal spin creates pairs; magnon-phonon modes
Sect. 14.2
measured response
Spin responses from neutron data
measured magnetic cross-section in place of a model
Sect. 14.4; Limit Check 14.3
break inversion and time reversal
Chiral, nonreciprocal targets
inversion and time reversal broken; cross responses
Sect. 14.6; Limit Check 14.7